Exam #1 - Out of Class Portion

Math 742 - Mathematical Statistics

2026-01-15

Exam Integrity Statement (Take-Home Portion)

This take-home exam is an individual assessment of your ability to translate statistical theory into working computational analysis. You may use R or Python and their standard libraries, but you may not use the internet, AI tools (e.g., ChatGPT, Copilot, WolframAlpha), or external code repositories (e.g., StackOverflow, GitHub) while completing this exam. All code and analysis must be your own work.

You may consult your own class notes and textbooks. The purpose of this exam is to evaluate your ability to independently implement, simulate, and interpret statistical methods. Submissions that show evidence of outside assistance or automated code generation will be treated as violations of academic integrity.

→ At the top of your .Rmd of .ipynb file you must write a sentence that states you have read and understand the exam integrety statement. ←

When you are finished with the exam, submit both the PDF and the .Rmd file or the .ipynb file to the dropbox on D2L.

Due Date: 2/13/2026 at 10:00am MST

For consistency, use the following seed value: 02112026. Do this at the beginning and do not reset it at any point during the exam.

Problem 6 (10 points) — Monte Carlo and Inequalities (10 points)

Let

$$X \sim \text{Exponential}(1).$$

(a) Using 10^6 simulated draws, estimate

$$E[X], \quad E[X^2], \quad (E[X])^2.$$

(b) Use your estimates to verify Jensen's inequality for $\phi(x) = x^2$.

(c) Estimate

$$P(|X - E[X]| > 2)$$

and compare it with the Chebyshev bound.

Problem 7 (10 points) — Transformations and Sampling Distributions (10 points)

Let

$$X_1, \ldots, X_n \sim \text{Exponential}(1), \qquad n = 50,$$

and define

$$T = \frac{1}{n} \sum_{i=1}^{n} X_i.$$

(a) Simulate 100,000 realizations of T and plot its empirical density.

(b) Overlay the asymptotic normal density predicted by the Central Limit Theorem.

(c) Compute and report the empirical bias and variance of T.

Problem 8 (10 points) — Likelihood, MLE, and Invariance (10 points)

Let

$$X_1, \ldots, X_n \sim \text{Poisson}(\lambda), \qquad n = 50,\ \lambda = 3.$$

(a) Simulate 10,000 data sets and compute the MLE

$$\hat{\lambda} = \bar{X}.$$

(b) Plot the sampling distribution of $\hat{\lambda}$ and compare it to its asymptotic normal distribution.

(c) Using invariance, compute

$$P\widehat{(X = 0)} = e^{-\hat{\lambda}}$$

and estimate its bias.